

90. Ethernet support for time synchronization protocols

90.1 Introduction

This clause specifies the optional Time Synchronization Service Interface (TSSI). The TSSI can be used to support protocols that require knowledge of packet egress and ingress time.

The TSSI is defined for the full-duplex mode of operation only. It supports MAC operation at various data rates. The MII (Clause 22), GMII (Clause 35), XGMII (Clause 46), 25GMII (Clause 106), XLGMII (Clause 81), CGMII (Clause 81), 50GMII (Clause 132), 200GMII (Clause 117), and 400GMII (Clause 117) specifications are all compatible with the generic Reconciliation Sublayer (gRS) defined in 90.5.

90.2 Overview

The goal of this clause is to provide an accurate indication of the transmission and reception initiation times of all packets, as required to support various time synchronization protocols, e.g., IEEE Std 1588-2008 [B41], and IEEE Std 802.1AS [B39].

The specific goals are to:

- a) Define the Time Synchronization Service Interface (TSSI), with associated service primitives;
- b) Add management registers to indicate the maximum and minimum data delays for estimation of link latency at the Time Synchronization Protocol (TimeSync) Client.

90.3 Relationship with other IEEE standards

Per 90.2, the TimeSync capability provides support for various time synchronization protocols, including e.g., IEEE Std 1588-2008 or IEEE Std 802.1AS. The definition of TimeSync Client, its capabilities and functions is outside the scope of IEEE Std 802.3.

90.4 Time Synchronization Service Interface (TSSI)

90.4.1 Introduction

This subclause specifies services provided by an extension to the Reconciliation Sublayers specified elsewhere in this standard.

90.4.1.1 Interlayer service interfaces

Figure 90–1 depicts the TimeSync Client and the RS interlayer service interfaces.

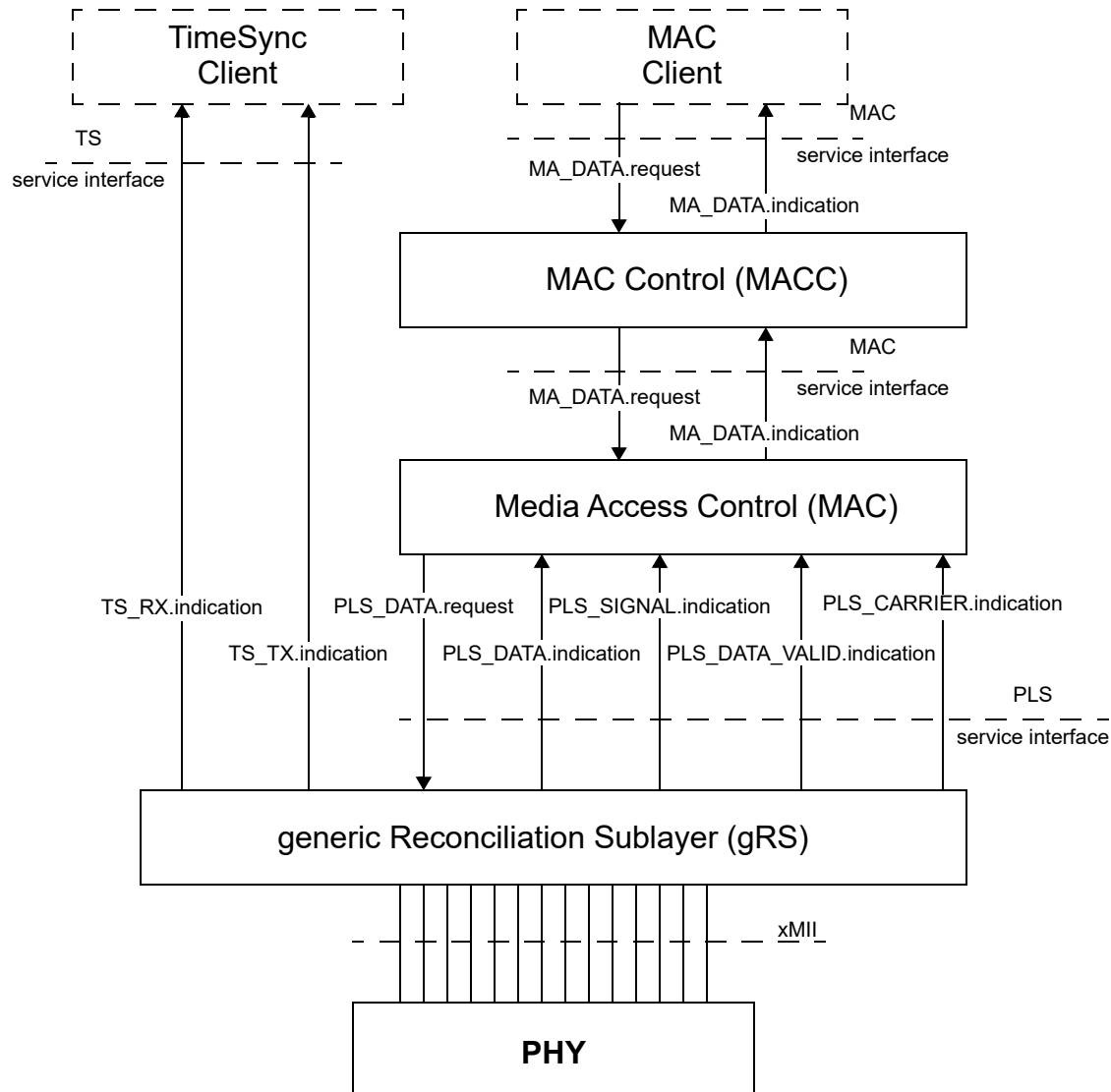


Figure 90–1—Relationship of the TimeSync Client, TSSI and gRS sublayer relative to MAC and MAC Client and associated interfaces

NOTE 1—In this figure, the xMII is used as a generic term for the Media Independent Interfaces for implementations of 10BASE-T1L, 10BASE-T1S, and 100 Mb/s and above. For example: for 100 Mb/s implementations this interface is called MII; for 1 Gb/s implementations, it is called GMII; for 10 Gb/s implementations, it is called XGMII; etc.

NOTE 2—Optional Low Power Idle (LPI) Client service interface not shown.

90.4.1.2 Responsibilities of TimeSync Client

The TimeSync Client can use the indication of the event corresponding to the egress and ingress of packets provided by the TSSI, combined with knowledge of the protocol frames, to select the egress and ingress times relevant to the protocol. Which frames are of interest to any particular protocol is beyond the scope of this standard.

The TimeSync Client can use the indication of the event corresponding to the egress and ingress of packets at the xMII provided by the TSSI, together with the information provided by the TimeSync PHY transmit data delay and TimeSync PHY receive data delay (see 30.13.1).

90.4.2 TSSI

The following specifies the service interface provided by the RS to the TimeSync Client. These services are described in an abstract manner and do not imply any particular implementation. The model used in this service specification is identical to that used in 1.2.2.

The following primitives are defined:

- TS_TX.indication
- TS_RX.indication

90.4.3 Detailed service specification

90.4.3.1 TS_TX.indication primitive

This primitive defines the transfer of the indication of an event between gRS and the TimeSync Client.

90.4.3.1.1 Semantics

The semantics of the primitive are as follows:

TS_TX.indication(SFD, MM)

The SFD parameter can take only one possible value, DETECTED. When asserted (SFD = DETECTED), the TimeSync Client is notified that a valid SFD was detected by the gRS sublayer TS_SFD_Detect_TX function (see 90.5.1) in the xMII transmit signals.

The MM parameter is mandatory when the MAC Merge sublayer (see Clause 99) is instantiated. The MM parameter, when present, can take one of two possible values, i.e., PMAC or EMAC. The value EMAC indicates the SMD-E (SFD) value has been detected at the xMII. The value PMAC indicates that an SMD-S value has been detected at the xMII (see Table 99–1). The MM parameter is not provided when MAC Merge sublayer is not instantiated.

90.4.3.1.2 Condition for generation

This primitive is generated by the gRS sublayer in response to detection of a valid SFD in the data stream transmitted across the xMII transmit signals. Specific conditions for generation of this primitive are described in 90.5.1.

90.4.3.1.3 Effect of receipt

The receipt of this primitive by the TimeSync Client is undefined.

90.4.3.2 TS_RX.indication primitive

This primitive defines the transfer of the indication of an event between gRS and the TimeSync Client.

90.4.3.2.1 Semantics

The semantics of the primitive are as follows:

TS_RX.indication(SFD, MM)

The SFD parameter can take only one possible value, DETECTED. When asserted (SFD = DETECTED), the TimeSync Client is notified that a valid SFD was detected by the gRS sublayer TS_SFD_Detect_RX function (see 90.5.2) in the xMII receive signals.

The MM parameter is mandatory when the MAC Merge sublayer (see Clause 99) is instantiated. The MM parameter, when present, can take one of two possible values, i.e., PMAC or EMAC. The value EMAC indicates the SMD-E (SFD) value has been detected at the xMII. The value PMAC indicates that an SMD-S value has been detected at the xMII (see Table 99–1). The MM parameter is not provided when MAC Merge sublayer is not instantiated.

90.4.3.2.2 Condition for generation

This primitive is generated by the gRS sublayer in response to detection of a valid SFD in the data stream received across the xMII receive signals. Specific conditions for generation of this primitive are described in 90.5.2.

90.4.3.2.3 Effect of receipt

The receipt of this primitive by the TimeSync Client is undefined.

90.5 generic Reconciliation Sublayer (gRS)

Within the scope of this clause, the term generic Reconciliation Sublayer (gRS) is used to denote any IEEE 802.3 Reconciliation Sublayer (RS) used to interface a MAC with any PHY supporting the TimeSync capability through the xMII. The behavior of the gRS depends on whether a MAC Merge sublayer (see Clause 99) is instantiated above the gRS.

For the purpose of the TimeSync capabilities, two new functions are defined in this subclause, namely TS_SFD_Detect_TX (see 90.5.1) and TS_SFD_Detect_RX (see 90.5.2), which are responsible for generation of the TS_TX.indication and TS_RX.indication service primitives, as defined in 90.4. Figure 90–2 presents the TS_SFD_Detect_TX and TS_SFD_Detect_RX functions and their location within the RS sublayer.

90.5.1 TS_SFD_Detect_TX function

The TS_SFD_Detect_TX function observes the xMII transmit signals.

When the MAC Merge sublayer is not instantiated, the TS_SFD_Detect_TX function detects the occurrence of the Start Frame Delimiter (SFD, see 3.1.1 and 3.2.2) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS_TX.indication shall be generated only when the SFD is detected on the transmit signals of the xMII (SFD=DETECTED).

When the MAC Merge sublayer is instantiated, the TS_SFD_Detect_TX function detects the occurrence of the Start mPacket Delimiter for an express packet or preemptable packet start (SMD-E or SMD-S, see 99.3.3) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS_TX.indication, shall be generated only when the SMD-E or SMD-S is detected on the transmit signals of the xMII (SFD=DETECTED). The value of MM shall indicate whether an SMD-E (MM=EMAC) or an SMD-S (MM=PMAC) was detected.

90.5.2 TS_SFD_Detect_RX function

The TS_SFD_Detect_RX function observes the xMII receive signals.

When the MAC Merge sublayer is not instantiated, the TS_SFD_Detect_RX function detects the occurrence of the Start Frame Delimiter (SFD, see 3.1.1 and 3.2.2) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS_RX.indication, shall be generated only when the SFD is detected on the receive signals of the xMII (SFD=DETECTED).

When the MAC Merge sublayer is instantiated, the TS_SFD_Detect_RX function detects the occurrence of the Start mPacket Delimiter for an express packet or preemptable packet start (SMD-E or SMD-S, see 99.3.3) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS_RX.indication, shall be generated only when the SMD-E or SMD-S is detected on the transmit signals of the xMII (SFD=DETECTED). The value of MM shall indicate whether an SMD-E (MM=EMAC) or an SMD-S (MM=PMAC) was detected.

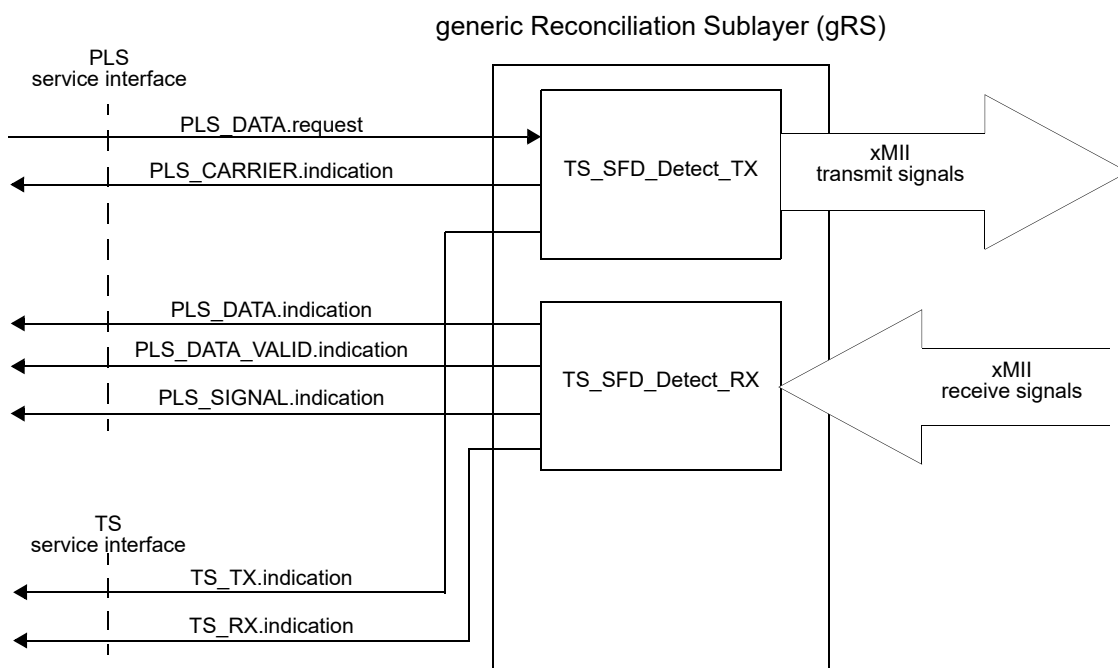


Figure 90-2—TS_SFD_Detect_TX and TS_SFD_Detect_RX functions within the generic Reconciliation Sublayer (gRS)

90.6 Overview of management features

Clause 30 describes the management functions for any of the IEEE Std 802.3 compliant PHYs. Objects defined in Clause 30 for the support of TimeSync capability are summarized below:

- oTimeSync managed object class, as defined in 30.13.1
- aTimeSyncCapabilityTX and aTimeSyncCapabilityRX managed objects, reflecting the status of a series of MDIO capability registers (1.1800, 2.1800, 3.1800, 4.1800, 5.1800, and 6.1800), as defined in 30.13.1.1 and 30.13.1.2, respectively
- aTimeSyncDelayTXmax and aTimeSyncDelayTXmin managed objects, representing the aggregate value of the series of transmit path data delay registers present in the instantiated MDIO registers (for each MMD, in case of multiple instances) 1.1801 through 1.1804, 2.1801 through 2.1804,

- 3.1801 through 3.1804, 4.1801 through 4.1804, 5.1801 through 5.1804, and 6.1801 through 6.1804, as defined in 30.13.1.3 and 30.13.1.4, respectively
- aTimeSyncDelayRXmax and aTimeSyncDelayRXmin managed objects, representing the aggregate value of the series of receive path data delay registers present in the instantiated MDIO registers (for each MMD, in case of multiple instances) 1.1805 through 1.1808, 2.1805 through 2.1808, 3.1805 through 3.1808, 4.1805 through 4.1808, 5.1805 through 5.1808, and 6.1805 through 6.1808, as defined in 30.13.1.5 and 30.13.1.6, respectively

The Management Data Input/Output (MDIO) capability described in Clause 45 defines several variables that provide TimeSync status information for the PMD, as shown in Table 90–1:

Table 90–1—Summary of TimeSync features in Clause 45

Register	Name	Reference
1.1800	TimeSync PMA/PMD capability register	45.2.1.175
1.1801 through 1.1804	TimeSync PMA/PMD transmit path data delay	45.2.1.176
1.1805 through 1.1808	TimeSync PMA/PMD receive path data delay	45.2.1.177
2.1800	TimeSync WIS capability register	45.2.2.20
2.1801 through 2.1804	TimeSync WIS transmit path data delay	45.2.2.21
2.1805 through 2.1808	TimeSync WIS receive path data delay	45.2.2.22
3.1800	TimeSync PCS capability register	45.2.3.67
3.1801 through 3.1804	TimeSync PCS transmit path data delay	45.2.3.68
3.1805 through 3.1808	TimeSync PCS receive path data delay	45.2.3.69
4.1800	TimeSync PHY XS capability register	45.2.4.28
4.1801 through 4.1804	TimeSync PHY XS transmit path data delay	45.2.4.29
4.1805 through 4.1808	TimeSync PHY XS receive path data delay	45.2.4.30
5.1800	TimeSync DTE XS capability register	45.2.5.28
5.1801 through 5.1804	TimeSync DTE XS transmit path data delay	45.2.5.29
5.1805 through 5.1808	TimeSync DTE XS receive path data delay	45.2.5.30
6.1800	TimeSync TC capability register	45.2.6.14
6.1801 through 6.1804	TimeSync TC transmit path data delay	45.2.6.15
6.1805 through 6.1808	TimeSync TC receive path data delay	45.2.6.16

90.7 Data delay measurement

The TimeSync capability requires measurement of data delay in the transmit and receive paths, as shown in Figure 90–3. The transmit path data delay is measured from the beginning of the SFD at the xMII input to the beginning of the SFD at the MDI output. The receive path data delay is measured from the beginning of the SFD at the MDI input to the beginning of the SFD at the xMII output.

For a PHY that includes an FEC function, the transmit and receive path data delays may show significant variation depending upon the position of the SFD within the FEC block. However, since the variation due to

this effect in the transmit path is expected to be compensated by the inverse variation in the receive path, it is recommended that the transmit and receive path data delays be reported as if the SFD is at the start of the FEC block.

The receiver of a multi-lane PHY is expected to include a buffer to compensate for skew between the lanes. This buffer selectively delays each lane such that the lanes are aligned at the buffer output. The earliest arriving lane experiences the most delay through the buffer and the latest arriving lane experiences the least delay through the buffer. The receive path data delay for a multi-lane PHY is reported as if the beginning of the SFD arrived at the MDI input on the lane with the smallest buffer delay.

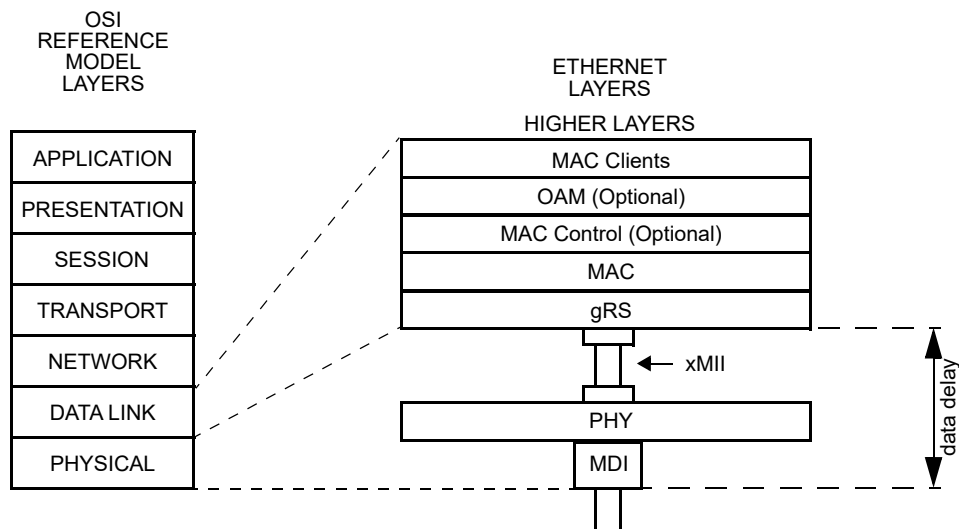


Figure 90–3—Data delay measurement

NOTE 1—The measured data delay in either the transmit or receive path should not include delay resulting from any length of the medium, regardless of the type of the medium used by the given PHY. It is observed that even if any medium-imposed delay is included in the measured data delay, it does not affect the operation of the TimeSync functions and only shifts the timing reference point farther from the MDI.

The obtained data delay measurement shall be reported in the form of a quartet of values; the maximum transmit data delay, the minimum transmit data delay, the maximum receive data delay, and the minimum receive data delay, as defined for the oTimeSync managed object class (30.13.1).

NOTE 2—The data delay values represent only the data delay in the PHY sublayers. The TimeSync Client may need to adjust for delays within the gRS. For example, the TimeSync Client may need to subtract the value of the delay associated with the TS_SFD_Detect_TX function from the sum of the minimum transmit data delay values reported by individual MMD(s). Likewise, the TimeSync Client may need to add the value of the delay associated with the TS_SFD_Detect_RX function to the sum of the maximum receive data delay values reported by individual MMD(s).

NOTE 3—For PHYs that are specified with an FEC sublayer separate from the PCS, the data delay for the FEC sublayer should be included in either the PCS delay registers or the PMA/PMD delay registers of the MMD in which the FEC sublayer is implemented, but not in both.

90.8 Protocol implementation conformance statement (PICS) proforma for Clause 90, Ethernet support for time synchronization protocols¹⁵⁹

90.8.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 90 Ethernet support for time synchronization protocols, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

90.8.2 Identification

90.8.2.1 Implementation identification

Supplier	
Contact point for inquiries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s)	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirements for the identification.	
NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

90.8.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 90, Ethernet support for time synchronization protocols
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022)	
Date of Statement	

¹⁵⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

90.8.3 TSSI indication

Item	Feature	Subclause	Value/Comment	Status	Support
*MM	MAC Merge	90.5	A MAC Merge sublayer is instantiated above gRS.	O	Yes [] No []
TS_TX	TS_TX.indication generation	90.5.1	TS_TX.indication is generated only when the SFD is detected on the transmit signals of the xMII.	!MM:M	Yes [] N/A []
TS_RX	TS_RX.indication generation	90.5.2	TS_RX.indication is generated only when the SFD is detected on the receive signals of the xMII.	!MM:M	Yes [] N/A []
TS_T2	TS_TX.indication generation with MAC Merge	90.5.1	TS_TX.indication is generated only when SMD-E or SMD-S is detected on the transmit signals of the xMII.	MM:M	Yes [] N/A []
TS_T3	MM parameter for TS_TX	90.5.1	Value of MM parameter indicates whether an SMD-E or SMD-S is detected.	MM:M	Yes [] N/A []
TS_R2	TS_RX.indication generation with MAC Merge	90.5.2	TS_RX.indication is generated only when SMD-E or SMD-S is detected on the receive signals of the xMII.	MM:M	Yes [] N/A []
TS_R3	MM parameter for TS_RX	90.5.2	Value of MM parameter indicates whether an SMD-E or SMD-S is detected.	MM:M	Yes [] N/A []

90.8.4 Data delay reporting

Item	Feature	Subclause	Value/Comment	Status	Support
Report	Data delay reporting format	90.7	Data delay measurement is reported in the form of a quartet of values; the maximum transmit data delay, the minimum transmit data delay, the maximum receive data delay, and the minimum receive data delay, for each MMD instantiated in the device.	M	Yes []